Software Requirements Specification

EswatiniHomes Property Listing System

Group Members: Simelane Njabulo -202301338
Mathonsi Simiso-202302294
Malaza Mncobi-202201551
Vilakati Noncedo-202402458

Standard Alignment: IEEE Std 830-1998

Contents

1 Introduction **2**
- 1.1 Purpose . . . 2
- 1.2 Scope . . . 2
- 1.3 Definitions, Acronyms, and Abbreviations . . . 2
- 1.4 References . . . 2
- 1.5 Overview . . . 2

2 Overall Description **3**
- 2.1 Product Perspective . . . 3
- 2.2 Product Functions . . . 3
- 2.3 User Characteristics . . . 3
- 2.4 Constraints . . . 3
- 2.5 Assumptions and Dependencies . . . 3

3 Specific Requirements **4**
- 3.1 External Interface Requirements . . . 4
  - 3.1.1 User Interfaces . . . 4
  - 3.1.2 Software Interfaces . . . 4
  - 3.1.3 Communications Interfaces . . . 4
- 3.2 Functional Requirements . . . 4
- 3.3 Performance Requirements . . . 4
- 3.4 Design Constraints . . . 4
- 3.5 Software System Attributes . . . 5
  - 3.5.1 Security . . . 5
  - 3.5.2 Reliability . . . 5
  - 3.5.3 Maintainability . . . 5
  - 3.5.4 Portability . . . 5

A Database Schema Overview **5**

1 Introduction

1.1 Purpose

This document specifies the requirements for the EswatiniHomes Property Listing System. It is intended for the developers, project managers, and the client (the real estate agency). The purpose of this SRS is to define the functional and non-functional requirements of the system, serving as a binding agreement between the stakeholders and the development team.

1.2 Scope

The software will be a web application that allows:

- **Guests:** Search, filter, and view property listings.
- **Agents:** Register, list properties, and track commissions/deals.
- **Admins:** Approve/reject listings, manage users, and view platform statistics.

The system will be built using HTML, CSS, PHP, and MySQL, and will run on an Apache web server (XAMPP for local development).

1.3 Definitions, Acronyms, and Abbreviations

SRS Software Requirements Specification

SDD Software Design Description

CRUD Create, Read, Update, Delete

CSRF Cross-Site Request Forgery

UAC User Account Control (Windows)

LAMP Linux, Apache, MySQL, PHP

1.4 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- XAMPP Documentation.
- EswatiniHomes Project Management Plan (SPMP).

1.5 Overview

This document is organized as follows: Section 2 provides an overall description of the product, including its perspective, functions, user characteristics, constraints, and assumptions. Section 3 contains the specific requirements, including external interfaces, functional requirements, performance, and software system attributes.

2 Overall Description

2.1 Product Perspective

This is a stand-alone web application built on the LAMP stack. It will run on a standard web browser and does not require any client-side installation. The system acts as a bridge between Agents (who provide property data) and Users (who consume property data).

2.2 Product Functions

The major functions of the system include:

- **User Authentication:** Login/Register for Admins, Agents, and standard Users.
- **Property Management:** Create, Read, Update, and Delete property listings.
- **Search & Filtering:** Search by location, property type, listing type, and max price.
- **Lead Generation:** Users can contact agents regarding specific properties.
- **Admin Dashboard:** Approve/Reject properties, manage users.
- **Agent Dashboard:** Track commissions and closed deals.

2.3 User Characteristics

- **Public Users:** Basic computer literacy. Expect a clean, mobile-responsive interface.
- **Agents:** Real estate professionals. Need efficient tools to upload images and track sales.
- **Admins:** Technical staff. Need full control over the database and site content.

2.4 Constraints

- **Database:** Must use MySQL/MariaDB.
- **Server:** Must run on Apache.
- **Security:** Passwords must be hashed (e.g., using PHP `password_hash()`).
- **Environment:** Must be compatible with PHP 7.4+ and PHP 8.x.

2.5 Assumptions and Dependencies

- The local development environment (XAMPP) is configured correctly (e.g., MySQL port changed to 3307 if 3306 is in use).
- The production server has the required PHP extensions enabled (e.g., `mysqli`, `fileinfo`).
- Users have a stable internet connection to load property images.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The system must be responsive and display correctly on desktop and mobile devices. The UI will use a green (#0f5132) and gold (#d4a017) color scheme. Key screens include:

- **Homepage:** Hero section, search bar, and a grid of property cards.
- **Login/Register:** Simple forms with validation.
- **Dashboards:** Sidebar navigation with data tables for managing properties and deals.

3.1.2 Software Interfaces

The system will interact with:

- **MySQL Database:** For storing all property, agent, user, and lead data.
- **Apache Web Server:** For serving PHP scripts and static assets.

3.1.3 Communications Interfaces

The system will communicate via HTTP/HTTPS protocols. Form data will be submitted via POST and GET methods.

3.2 Functional Requirements

3.3 Performance Requirements

- **Load Time:** The homepage *shall* load in under 2 seconds on a standard broadband connection.
- **Concurrency:** The system *shall* support at least 50 concurrent users without database timeout.
- **Pagination:** The property grid *shall* display a maximum of 12 properties per page to optimize load times.

3.4 Design Constraints

- Must use a relational database (MySQL).
- Must use the `mysqli` extension for database interactions.
- The application must run on a standard XAMPP environment without requiring external server configurations.

ID	Requirement Description
FR-01	The system *shall* allow agents to upload an image (JPG, PNG) for a property.
FR-02	The system *shall* allow users to filter properties by a price range.
FR-03	The system *shall* prevent unauthorized users from accessing the admin dashboard.
FR-04	A property *shall not* appear on the public homepage until an Admin changes its status to 'Approved'.
FR-05	When a property is deleted, the system *shall* also delete the associated image file from the server.
FR-06	The system *shall* use CSRF tokens to validate all POST requests.
FR-07	The system *shall* use Prepared Statements for all database queries to prevent SQL Injection.
FR-08	The system *shall* calculate and display agent commissions based on deal amounts.
FR-09	The system *shall* display "No Image Available" if a property image is missing.
FR-10	The system *shall* allow Admins to delete users and properties.

Table 1: Functional Requirements Matrix

3.5 Software System Attributes

3.5.1 Security

- All passwords must be encrypted using `password_hash()`.
- All user inputs must be sanitized using `htmlspecialchars()` to prevent XSS attacks.
- The `delete.php` script must require a valid session and CSRF token.

3.5.2 Reliability

The system should handle database connection failures gracefully, displaying a user-friendly error message rather than exposing raw SQL errors.

3.5.3 Maintainability

The code must be modular. Database connections, helper functions (e.g., `currency()`), and authentication logic must be separated into an `include/config.php` file.

3.5.4 Portability

The system must be compatible with any web server running PHP 7.4+ and MySQL 5.7+.

A Database Schema Overview

The system uses a relational database named `property_listing`. The core tables are:

- `properties` (id, title, description, price, location, status, agent_id)
- `agents` (id, name, email, phone)
- `users` (id, username, password, role)
- `leads` (id, property_id, name, message)
- `agent_deals` (id, agent_id, deal_amount, commission_amount)